BESS OPERATIONS REPORT

April 2026

Site:	Bukhara BESS
Reporting period:	01 April 2026 — 30 April 2026
Blocks monitored:	15 blocks
Report generated:	2026-05-16 11:02
Report type:	Monthly Operations & Performance Summary

Report generated automatically from SCADA exports. Per-block KPIs use the validated availability engine. RTE / EFC / throughput sourced from cumulative energy counters.

Executive Summary

77.9%	92.3%	6,929	47.0
Fleet Availability	Fleet RTE	Discharge (MWh)	Avg EFC / block

15	3,980	13,104	1.9
Blocks monitored	Production events	Persistent warnings	Excluded days (avg)

During **01 April 2026 — 30 April 2026**, the Bukhara BESS fleet (15 blocks) maintained an availability of **77.91%** (below the 95% target) and a round-trip efficiency of **92.27%** (above the 85% target). Fleet throughput totalled **7,509 MWh charge** and **6,929 MWh discharge**, equivalent to **705 full cycles** across the site (avg 47.0 per block). **3,980** production-impacting events were recorded after deduplication (24,460 additional transient warnings were filtered as noise). **3** block(s) flagged as underperforming versus fleet mean.

Site Overview

Parameter	Value
Site	Bukhara BESS
Number of blocks	15
Reporting period	01 April 2026 — 30 April 2026
Excluded days (fleet, avg)	1.9 day(s) — quality flag EXCLUDED
Total fleet charge	7,509.5 MWh
Total fleet discharge	6,929.1 MWh
Throughput	14,438.6 MWh
Fleet EFC (cycle counter delta)	704.8 cycles

Block Availability

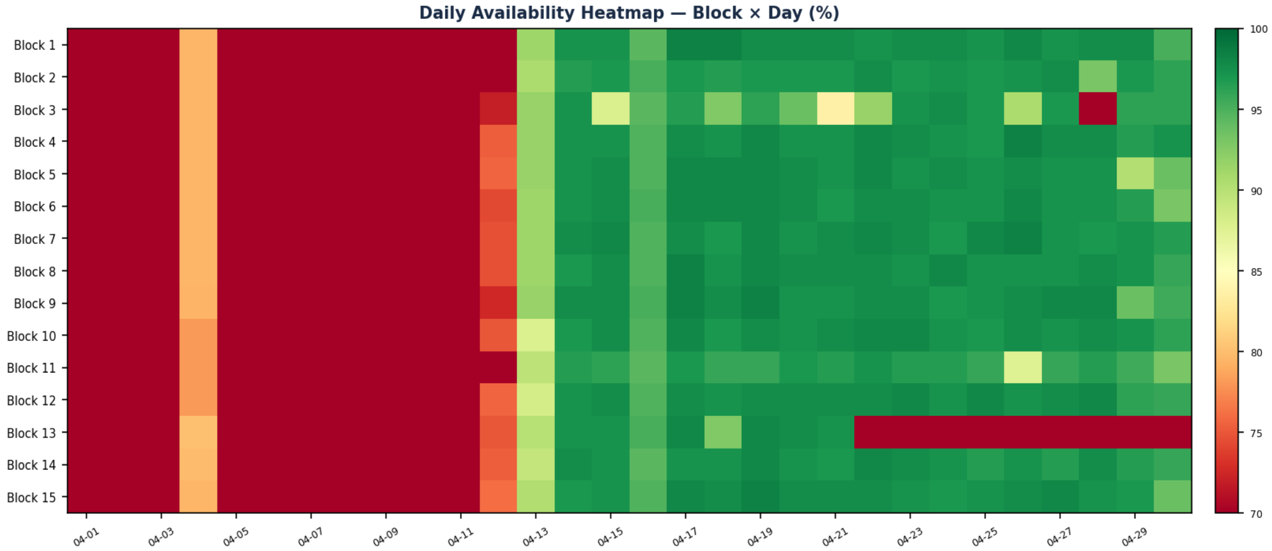

Figure 1: Daily availability per block. Red cells indicate planned or unplanned outage; green cells indicate full operation.

Block	Avg Availability	Excluded Days	Status
Block 13.0	65.15%	2	Low
Block 11.0	71.05%	1	Low
Block 3.0	72.58%	2	Low
Block 2.0	75.32%	2	Low
Block 4.0	75.80%	2	Low
Block 1.0	76.06%	2	Low
Block 7.0	76.23%	2	Low
Block 6.0	76.25%	2	Low
Block 5.0	76.44%	2	Low
Block 10.0	76.74%	1	Low
Block 9.0	76.83%	2	Low
Block 12.0	77.00%	2	Low
Block 15.0	77.00%	2	Low
Block 14.0	77.03%	2	Low
Block 8.0	77.13%	2	Low

■ Energy & Round-Trip Efficiency

7,509	6,929	92.27%	14,439
Total Charge (MWh)	Total Discharge (MWh)	Fleet RTE	Total Throughput (MWh)

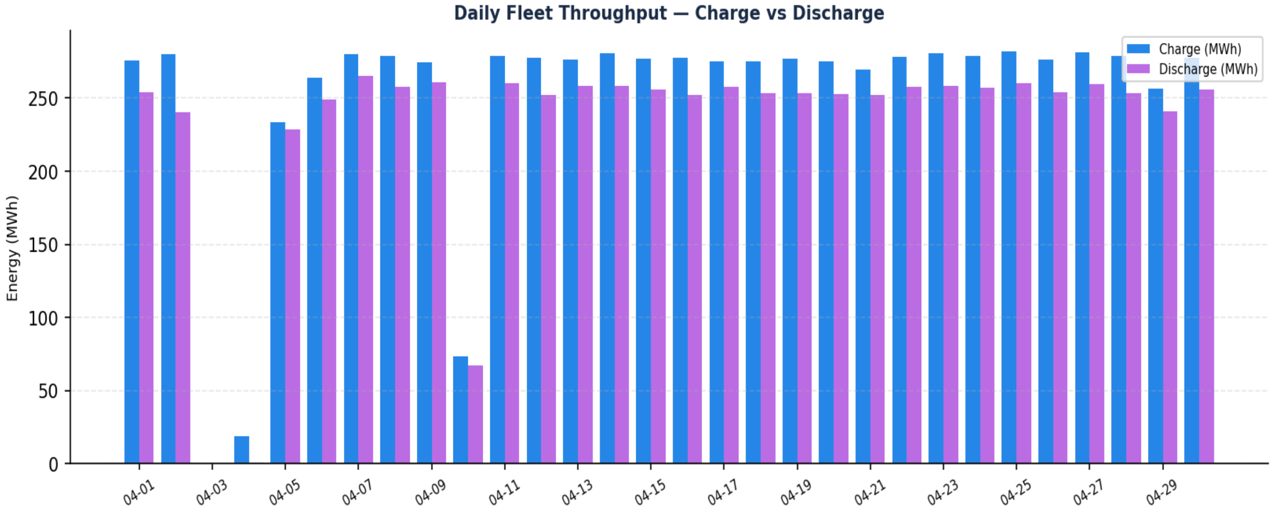

Figure 2: Daily fleet throughput (charge vs discharge)

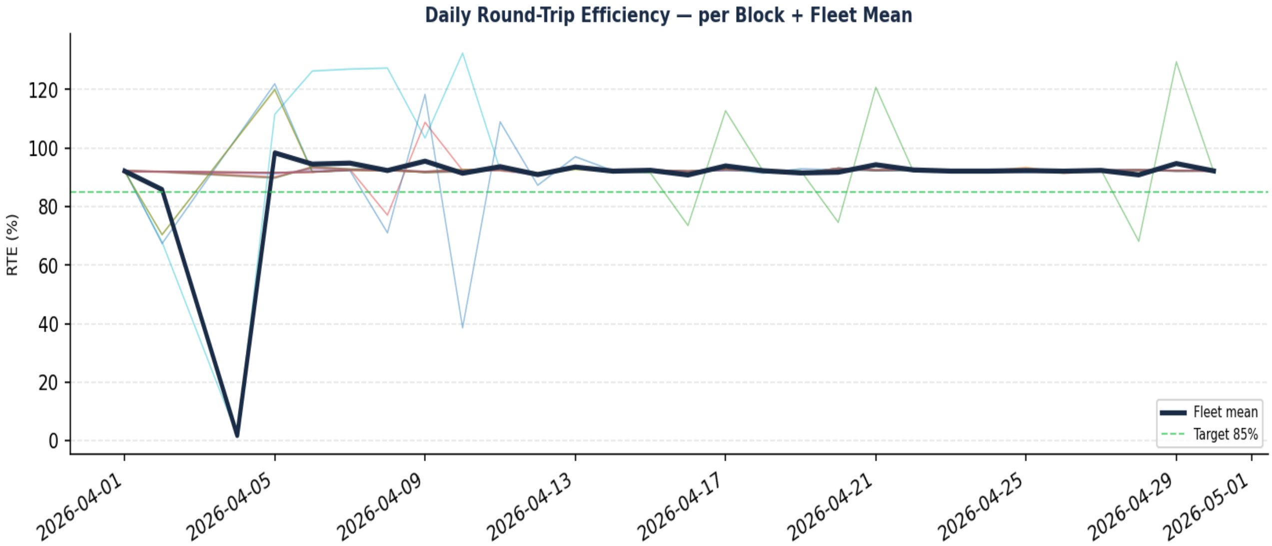

Figure 3: Daily RTE per block (light lines) and fleet mean (dark line). Each block-day uses RTE = discharge / charge from cumulative counters. Days flagged as EXCLUDED or INCOMPLETE are omitted.

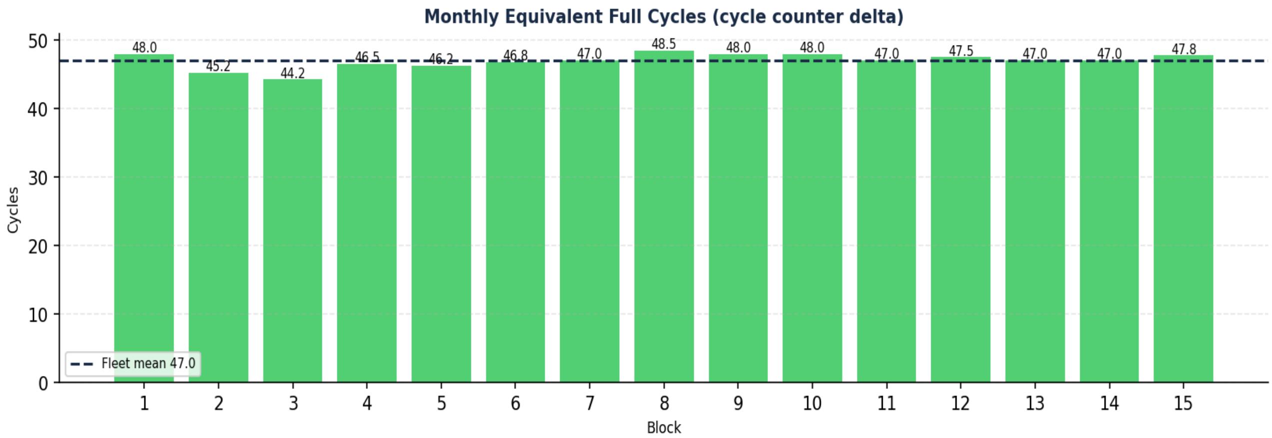

Figure 4: Monthly EFC per block — delta of the cycle counter between start and end of period (sourced directly from Battery Unit data).

■ Daily Block Performance

Per-block daily KPI table — one row per (date, block). Quality flag: **VALID** when day has full data and 50% < RTE < 100%; **INCOMPLETE** when throughput < 100 kWh; **EXCLUDED** when both charge and discharge are zero (site offline). Showing first 30 and last 5 rows (full table in appendix CSV).

Date	Block	Chg MWh	Dis MWh	RTE	EFC	Avail	Op h	Alarms	Flag
2026-04-01	Block 1	18.80	17.31	92.1%	1.60	43.2%	9.8	10	VALID
2026-04-01	Block 2	18.38	16.94	92.2%	1.51	42.7%	9.5	10	VALID
2026-04-01	Block 3	18.87	17.40	92.2%	1.48	43.1%	9.8	10	VALID
2026-04-01	Block 4	18.80	17.32	92.1%	1.55	43.1%	9.7	10	VALID
2026-04-01	Block 5	18.80	17.31	92.1%	1.54	43.5%	9.8	10	VALID
2026-04-01	Block 6	18.75	17.29	92.2%	1.56	49.0%	9.5	19	VALID
2026-04-01	Block 7	14.10	12.93	91.7%	1.57	28.5%	4.6	118	VALID
2026-04-01	Block 8	18.82	17.34	92.2%	1.62	42.7%	9.6	10	VALID
2026-04-01	Block 9	18.79	17.31	92.1%	1.60	44.1%	9.6	29	VALID
2026-04-01	Block 10	18.74	17.27	92.2%	1.60	42.7%	9.6	10	VALID
2026-04-01	Block 11	18.14	16.78	92.5%	1.57	41.0%	8.9	10	VALID
2026-04-01	Block 12	18.63	17.12	91.9%	1.58	47.9%	9.2	43	VALID
2026-04-01	Block 13	18.70	17.23	92.2%	1.57	42.0%	9.5	10	VALID
2026-04-01	Block 14	18.75	17.28	92.1%	1.57	42.0%	9.5	10	VALID
2026-04-01	Block 15	18.82	17.33	92.1%	1.59	43.4%	9.8	10	VALID
2026-04-02	Block 1	18.72	17.20	91.9%	1.60	37.5%	8.6	14	VALID
2026-04-02	Block 2	18.36	16.86	91.8%	1.51	37.6%	8.3	14	VALID
2026-04-02	Block 3	18.82	17.30	91.9%	1.48	36.8%	8.6	20	VALID
2026-04-02	Block 4	18.75	17.24	92.0%	1.55	37.1%	8.5	20	VALID
2026-04-02	Block 5	18.68	17.18	92.0%	1.54	38.2%	8.6	16	VALID
2026-04-02	Block 6	18.82	17.25	91.7%	1.56	35.4%	8.2	18	VALID
2026-04-02	Block 7	18.78	17.26	91.9%	1.57	37.5%	8.6	21	VALID
2026-04-02	Block 8	18.75	17.23	91.9%	1.62	36.8%	8.5	20	VALID
2026-04-02	Block 9	18.70	17.18	91.9%	1.60	37.6%	8.5	21	VALID
2026-04-02	Block 10	18.66	12.67	67.9%	1.60	30.9%	7.2	18	VALID
2026-04-02	Block 11	18.37	12.38	67.3%	1.57	31.2%	7.2	18	VALID
2026-04-02	Block 12	18.54	13.06	70.4%	1.58	33.3%	7.7	20	VALID
2026-04-02	Block 13	18.64	13.12	70.4%	1.57	33.0%	7.6	22	VALID
2026-04-02	Block 14	18.63	17.14	92.0%	1.57	37.5%	8.5	20	VALID
2026-04-02	Block 15	18.67	17.17	92.0%	1.59	37.5%	8.5	19	VALID

■ Block Performance Anomalies

3 block(s) flagged as underperforming (RTE z-score < -1 and/or throughput z-score < -1):

Block	Avg RTE	RTE z	Avg Throughput	Th. z	Avg Avail	Total Alarms
Block 3	89.68%	-2.57	33.0 MWh	-2.43	72.86%	206
Block 11	90.33%	-1.74	34.5 MWh	+0.06	80.12%	129
Block 5	92.18%	+0.61	33.9 MWh	-1.02	77.70%	212

■ Alarm & Fault Analysis

3,980	13,104	24,460	1,268
Production (dedup)	Persistent warnings	Transient (filtered)	Echo duplicates

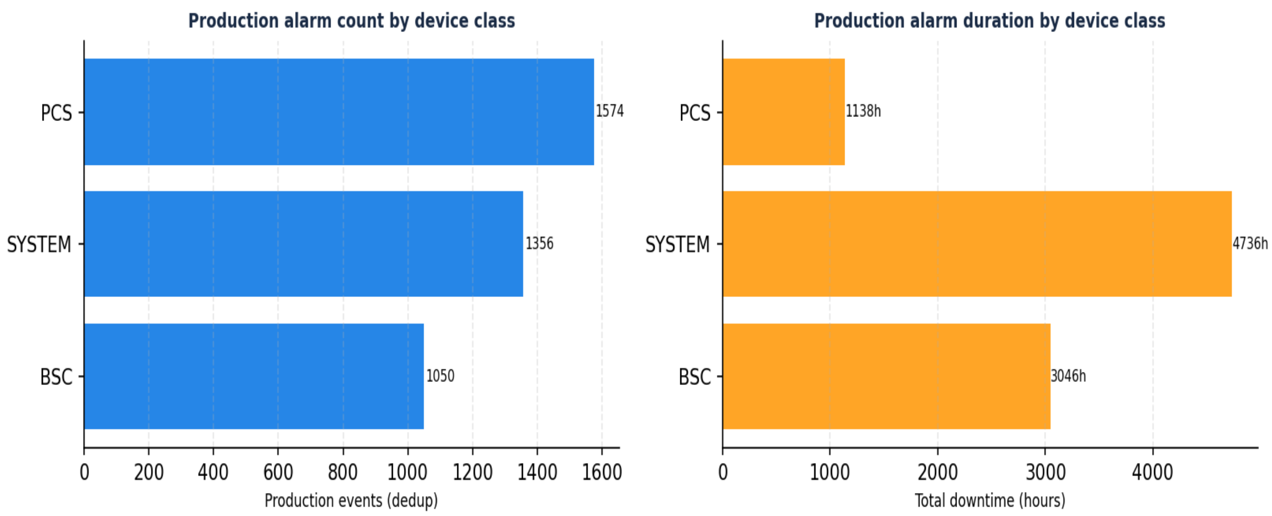

Figure 5: Production alarms grouped by device class. Counts (left) and total downtime (right). Echo duplicates removed; transient warnings < 1 min excluded.

Top 10 Production Trigger Names (deduplicated)

Trigger Name	Count
Inverter In Key Stop	1356
BSC UNIT FAULT	667
LC - SYSTEM FAULT STATUS1 ACTIVE	579
LC-PCS COMM FAULT	402
BSC FSS FAULT	383
LC-BSC FAULT	363
PCS AC SWITCH FAULT	112
PCS SOFT START FAULT	75

Trigger Name	Count
PCS ISLAND PROTECTION	23
LC-BSC COMM FAULT	6

Top 10 Longest Production Events (sorted by duration)

Activated	Element	Trigger	Duration
2026-04-02 20:34	System Info 13	LC - SYSTEM FAULT STATUS1 ACTIVE	46.6 h
2026-04-02 20:34	System Info 12	LC - SYSTEM FAULT STATUS1 ACTIVE	46.6 h
2026-04-02 20:55	BSC 13.02	BSC UNIT FAULT	46.3 h
2026-04-02 20:55	BSC 13.01	BSC UNIT FAULT	46.3 h
2026-04-02 20:55	System Info 13	LC-PCS COMM FAULT	46.3 h
2026-04-02 20:55	System Info 12	LC-PCS COMM FAULT	46.3 h
2026-04-02 20:55	System Info 13	LC-BSC FAULT	46.3 h
2026-04-02 20:34	System Info 11	LC - SYSTEM FAULT STATUS1 ACTIVE	46.2 h
2026-04-02 20:34	System Info 10	LC - SYSTEM FAULT STATUS1 ACTIVE	46.2 h
2026-04-02 20:55	BSC 12.02	BSC UNIT FAULT	45.8 h

■ Conclusions & Observations

1. Fleet availability of **77.91%** was recorded (below the 95% target). Fleet RTE was **92.27%** (above the 85% target).

2. Across 15 blocks, the fleet delivered **6,929.1 MWh** of discharge energy on **7,509.5 MWh** of charge energy, equivalent to **705** equivalent full cycles (avg 47.0 per block).

3. Fleet-wide average of **1.9** days per block were flagged as EXCLUDED (zero throughput on both sides — likely site outage or SCADA export gap). These days are removed from RTE/availability calculations.

4. After dedup and transient filtering, **3,980** production-impacting events were recorded (plus 13,104 persistent warnings; 24,460 transient events filtered as noise).

5. **3** block(s) flagged as underperforming versus fleet mean (RTE z-score < -1 or throughput z-score < -1).